

Homework 1: The Relational Model

Name:

Instructions: Print this assignment using single-side pages. Write in the solutions in the space provided below each question. You are allowed to use the back of each page. If you used any scratch paper to show your work, append those to the end. **Submission:** Scan all pages into a PDF and submit on canvas.

Problems

1. Before the relational model was established, most organizations organized their data using the tree or hierarchical model. Suppose a hospital stores patient information using the following hierarchy: Patient → Visit → Doctor. For example:

```
Adam Smith
  Visit 1
    Dr. Lee
  Visit 2
    Dr. Gomez
America Chambers
  Visit 1
    Dr. Lee
```

Answer the following questions.

- (a) The hospital begins recording medical conditions. A patient can have several conditions, and many patients can have the same condition. Where should conditions appear in the hierarchy? Explain the difficulty.
- (b) The hierarchy is not incorrect. So what is the actual weakness of using it as the hospitals general-purpose data model?

2. Consider the relation: **Course**(Dept, CourseNum, Title, Credits)

Assume that course numbers are unique only within a department. For example, both CS and MATH may have a course numbered 101.

(a) Is $\{\text{CourseNum}\}$ a superkey? Explain.

(b) List all superkeys.

(c) Is $\{\text{Dept}, \text{CourseNum}\}$ a candidate key? Explain.

3. Consider the relation: **Enrollment**(StudentID, CourseID, Semester, Grade)

(a) Why is $\{\text{StudentID}, \text{CourseID}\}$ not necessarily a key?

(b) Give a reasonable candidate key for this relation.

(c) Explain why $\{\text{StudentID}, \text{CourseID}, \text{Semester}, \text{Grade}\}$ is a superkey but not a candidate key.

4. Given a relational schema $R(A_1, A_2, \dots, A_n)$, what is the minimum and maximum number of superkeys that R can have? Explain.
5. Given a relational schema $R(A_1, A_2, \dots, A_n)$, what is the maximum number of relation instances, *not tuples*, that R can have? Explain your answer. (Hint: Recall that the domain $dom(A)$ of an attribute A is defined to be the set of legal values of A).

6. Consider the following relations for a database that keeps track of auto sales from a car dealership.

<code>car(vin, model, make, basePrice)</code>	<code>options(vin, optionItem, price)</code>
<code>sales(salespersonID, vin, date, price)</code>	<code>salesperson(name, email)</code>

Here are some assumptions to go on:

- Dealerships often have multiple **cars** that have the same make (e.g., Toyota) and model (e.g., Prius), so cars can only be identified by their vehicle identification number, or vin. Each car also has a base price, which does not include any options that can be added to it.
- Some cars can be equipped with additional **options** (such as a sunroof, window tints, or LED bulbs). Each option carries a different price for each car.
- Each **salesperson** can be uniquely identified by their email address.
- To record a **sale**, we store references to the salesperson, the car, date, and the price that it ultimately negotiated and sold for. Note that the same car may be sold more than once, even by the same salesperson. (This happens when the car is traded-in, or is sold back to the dealership at a later date.)
- To avoid scams, the dealership has a standing policy that prohibits the same car to be sold more than once on the same day.

Draw the schema diagram of this dealership's database. Each relation is represented by a box, with its attributes listed within. Underline the primary-key attributes in each relation. Add directed edges indicating foreign-key relationships.

7. The car dealership from the previous question hired a consulting group to optimize their database. The consultants suggested an alternate database schema to use instead. They collapsed all the data down to only two tables. Critique this new schema. Compared to previous schema, list as many problems as you can spot.

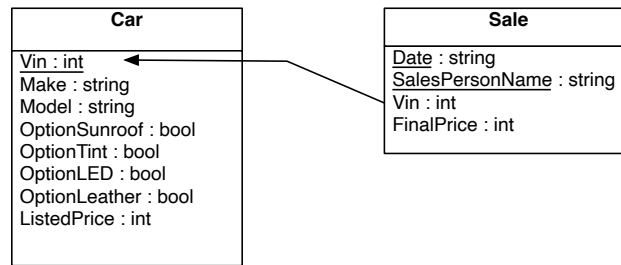


Figure 1: Newly Proposed Schema